UNIT IV

CORRELATION, REGRESSION AND STATISTICAL QUALITY CONTROL

Correlation | Regression | $\bar{X}$, R, p, np and c Charts

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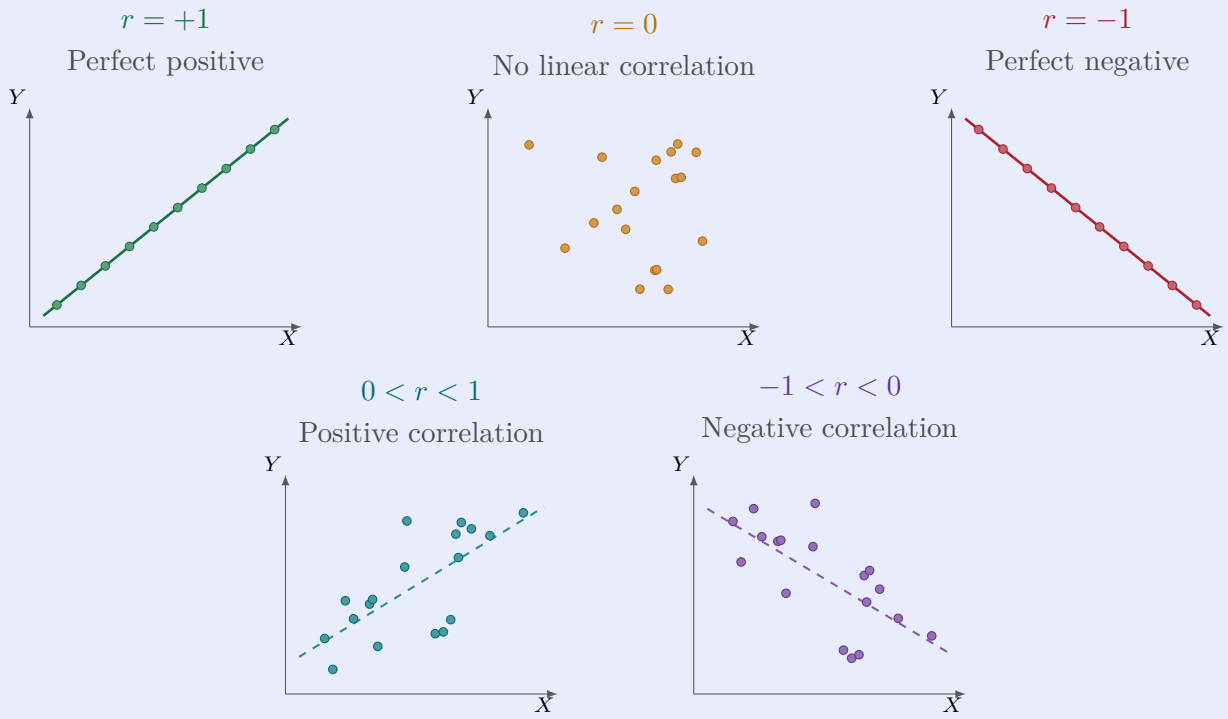
1 Correlation and Rank Correlation

Karl Pearson's Coefficient of Correlation (r)

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$$

- Measures the linear relationship between two variables; $-1 \le r \le 1$.
- $r = +1$: perfect positive correlation; $r = -1$: perfect negative correlation; $r = 0$: no linear correlation.
- Change of origin and scale does not affect r.

Scatter diagrams for different values of r



The closer the points lie to a straight line, the closer $|r|$ is to 1; an upward trend gives $r > 0$ and a downward trend gives $r < 0$.

Supporting summations

$$\sum x, \qquad \sum y, \qquad \sum x^2, \qquad \sum y^2, \qquad \sum xy$$

These five values are substituted directly in the correlation formula.

Spearman's Rank Correlation Coefficient (ρ)

$$\rho = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

- Used when the data is in rank form; $-1 \le \rho \le 1$.

- d = difference between the pair of ranks; n = number of ranked items.

Repeated ranks – correction factor (C.F)

If an item is repeated, assign the average of the repeated rank positions and use

$$\rho = 1 - \frac{6\,[\sum d^2 + C.F]}{n(n^2-1)}, \qquad C.F = \sum \frac{m\,(m^2-1)}{12}$$

where m is the number of times an item is repeated (one term for each repeated item).

Interpretation of r and ρ

Magnitude	Interpretation
0.00 to ±0.30	Weak correlation
±0.30 to ±0.70	Moderate correlation
±0.70 to ±1.00	Strong correlation

1.1 Problem 1 – Sales and expenditure

Problem 1

Find the coefficient of correlation between Sales (x) and Expenditure (y) for the following 10 items.

Sales (x)	50	50	55	60	65	65	65	60	60	50
Expenditure (y)	11	13	14	16	16	15	15	14	13	13

Solution – Table of summations ($n = 10$)

x	y	x^2	y^2	xy
50	11	2500	121	550
50	13	2500	169	650
55	14	3025	196	770
60	16	3600	256	960
65	16	4225	256	1040
65	15	4225	225	975
65	15	4225	225	975
60	14	3600	196	840
60	13	3600	169	780
50	13	2500	169	650
$\sum x = 580$	$\sum y = 140$	$\sum x^2 = 34000$	$\sum y^2 = 1982$	$\sum xy = 8190$

Karl Pearson's coefficient of correlation

$$
\begin{aligned}
r &= \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}} \\
&= \frac{10(8190) - (580)(140)}{\sqrt{[10(34000) - (580)^2]\,[10(1982) - (140)^2]}} \\
&= \frac{81900 - 81200}{\sqrt{[340000 - 336400]\,[19820 - 19600]}} \\
&= \frac{700}{\sqrt{3600 \times 220}} \\
&= \frac{700}{\sqrt{792000}} \\
&= \frac{700}{889.94} \\
r &= 0.7866 \qquad \Rightarrow \qquad \boxed{r \approx 0.79}
\end{aligned}
$$

Conclusion

$r \approx 0.79$ **: high positive correlation between sales and expenditure.**

1.2 Problem 2 – Weight and blood pressure

Problem 2

A study has been conducted to find whether there is any relationship between the weight and the blood pressure of an individual. Determine the coefficient of correlation for the following data.

Weight x (kg)	78	86	72	82	80	88	84	89	68	71
B.P. y (mm)	140	160	134	144	180	176	174	178	128	132

Solution – Table of summations ($n = 10$)

x	y	x^2	y^2	xy
78	140	6084	19600	10920
86	160	7396	25600	13760
72	134	5184	17956	9648
82	144	6724	20736	11808
80	180	6400	32400	14400
88	176	7744	30976	15488
84	174	7056	30276	14616
89	178	7921	31684	15842
68	128	4624	16384	8704
71	132	5041	17424	9372
$\sum x = 798$	$\sum y = 1546$	$\sum x^2 = 64174$	$\sum y^2 = 243036$	$\sum xy = 124558$

Karl Pearson's coefficient of correlation

$$
\begin{aligned}
r &= \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}} \\
&= \frac{10(124558) - (798)(1546)}{\sqrt{[10(64174) - (798)^2]\,[10(243036) - (1546)^2]}} \\
&= \frac{1245580 - 1233708}{\sqrt{[641740 - 636804]\,[2430360 - 2390116]}} \\
&= \frac{11872}{\sqrt{4936 \times 40244}} \\
&= \frac{11872}{\sqrt{198644384}} \\
&= \frac{11872}{14094.13}
\end{aligned}
$$

$r = 0.8423 \quad \Rightarrow \quad \boxed{r \approx 0.84}$

Conclusion

$r \approx 0.84$ **: strong positive correlation between weight and blood pressure.**

1.3 Problem 3 – Rank correlation (no repeated ranks)

Problem 3

Find the rank correlation coefficient for the following data.

Rank in x (R_x)	1	2	3	4	5	6	7
Rank in y (R_y)	4	3	1	2	6	5	7

Solution – Table of rank differences ($n = 7$)

R_x	R_y	$d = R_x - R_y$	d^2
1	4	-3	9
2	3	-1	1
3	1	2	4
4	2	2	4
5	6	-1	1
6	5	1	1
7	7	0	0
			$\sum d^2 = 20$

Spearman's rank correlation coefficient

$$\rho = 1 - \frac{6\sum d^2}{n\,(n^2-1)}$$

Here $n = 7, \qquad \sum d^2 = 20$

$$\rho = 1 - \frac{6(20)}{7\,(7^2-1)}$$

$$= 1 - \frac{120}{7 \times 48}$$

$$= 1 - \frac{120}{336}$$

$$= 1 - 0.3571$$

$$\rho = 0.6429 \qquad \Rightarrow \qquad \boxed{\rho \approx 0.64}$$

1.4 Problem 4 – Rank correlation with repeated ranks

Problem 4

Find the rank correlation coefficient for the following marks.

Marks in x	68	84	75	50	64	80	75	40	55	64
Marks in y	62	58	68	45	81	60	68	48	50	70

Assigning ranks (smallest mark = rank 1)

Sorted x: $40, 50, 55, 64, 64, 68, 75, 75, 80, 84 \Rightarrow 40 \to 1,\ 50 \to 2,\ 55 \to 3,\ 64, 64 \to \frac{4+5}{2} = 4.5,\ 68 \to 6,\ 75, 75 \to \frac{7+8}{2} = 7.5,\ 80 \to 9,\ 84 \to 10.$

Sorted y: $45, 48, 50, 58, 60, 62, 68, 68, 70, 81 \Rightarrow 45 \to 1,\ 48 \to 2,\ 50 \to 3,\ 58 \to 4,\ 60 \to 5,\ 62 \to 6,\ 68, 68 \to \frac{7+8}{2} = 7.5,\ 70 \to 9,\ 81 \to 10.$

Solution – Table of rank differences ($n = 10$)

x	y	R_x	R_y	$d = R_x - R_y$	d^2
68	62	6	6	0	0
84	58	10	4	6	36
75	68	7.5	7.5	0	0
50	45	2	1	1	1
64	81	4.5	10	−5.5	30.25
80	60	9	5	4	16
75	68	7.5	7.5	0	0
40	48	1	2	−1	1
55	50	3	3	0	0
64	70	4.5	9	−4.5	20.25
					$\sum d^2 = 104.5$

Correction factor (C.F) for repeated ranks

m = number of times an item is repeated.

- In x : 64 is repeated 2 times $\Rightarrow m = 2$; 75 is repeated 2 times $\Rightarrow m = 2$.
- In y : 68 is repeated 2 times $\Rightarrow m = 2$.

$$
\begin{aligned}
C.F &= \sum \frac{m\left(m^2-1\right)}{12} \\
&= \frac{2\left(2^2-1\right)}{12} + \frac{2\left(2^2-1\right)}{12} + \frac{2\left(2^2-1\right)}{12} \\
&= \frac{6}{12} + \frac{6}{12} + \frac{6}{12} \\
&= 0.5 + 0.5 + 0.5 \\
C.F &= 1.5
\end{aligned}
$$

Spearman's rank correlation coefficient with C.F

$$\rho = 1 - \frac{6\left[\sum d^2 + C.F\right]}{n\left(n^2-1\right)}$$

Here $n = 10, \qquad \sum d^2 = 104.5, \qquad C.F = 1.5$

$$
\begin{aligned}
\rho &= 1 - \frac{6\left[104.5 + 1.5\right]}{10\left(10^2-1\right)} \\
&= 1 - \frac{6(106)}{10 \times 99} \\
&= 1 - \frac{636}{990} \\
&= 1 - 0.6424 \\
\rho &= 0.3576 \qquad \Rightarrow \qquad \boxed{\rho \approx 0.36}
\end{aligned}
$$

Conclusion

$\rho \approx 0.36$ **: low positive rank correlation between the marks in** x **and** y**.**

1.5 Problem 5 – Musical contest rankings

Problem 5

Ten competitors in a musical contest were ranked by three judges A, B, C in the following order. Using Spearman's rank correlation, find which pair of judges have the same liking in music.

Rank by A	1	6	5	10	3	2	4	9	7	8
Rank by B	3	5	8	6	4	7	10	2	1	9
Rank by C	4	6	9	8	1	2	3	10	5	7

Solution – Table of rank differences ($n = 10$)

$$d_1 = R_A - R_B, \qquad d_2 = R_A - R_C, \qquad d_3 = R_B - R_C$$

R_A	R_B	R_C	d_1	d_1^2	d_2	d_2^2	d_3	d_3^2
1	3	4	-2	4	-3	9	-1	1
6	5	6	1	1	0	0	-1	1
5	8	9	-3	9	-4	16	-1	1
10	6	8	4	16	2	4	-2	4
3	4	1	-1	1	2	4	3	9
2	7	2	-5	25	0	0	5	25
4	10	3	-6	36	1	1	7	49
9	2	10	7	49	-1	1	-8	64
7	1	5	6	36	2	4	-4	16
8	9	7	-1	1	1	1	2	4
				$\sum d_1^2 = 178$		$\sum d_2^2 = 40$		$\sum d_3^2 = 174$

(i) Judges A and B

$$\begin{aligned}
\rho_{AB} &= 1 - \frac{6\sum d_1^2}{n\,(n^2-1)} \\
&= 1 - \frac{6(178)}{10\,(10^2-1)} \\
&= 1 - \frac{1068}{990} \\
&= 1 - 1.0788 \\
\rho_{AB} &= -0.0788 \quad \Rightarrow \quad \boxed{\rho_{AB} \approx -0.08}
\end{aligned}$$

(ii) Judges A and C

$$\begin{aligned}
\rho_{AC} &= 1 - \frac{6\sum d_2^2}{n\,(n^2-1)} \\
&= 1 - \frac{6(40)}{10\,(10^2-1)} \\
&= 1 - \frac{240}{990} \\
&= 1 - 0.2424 \\
\rho_{AC} &= 0.7576 \quad \Rightarrow \quad \boxed{\rho_{AC} \approx 0.76}
\end{aligned}$$

(iii) Judges B and C

$$\begin{aligned}\rho_{BC} &= 1 - \frac{6\sum d_3^2}{n\,(n^2-1)} \\ &= 1 - \frac{6(174)}{10\,(10^2-1)} \\ &= 1 - \frac{1044}{990} \\ &= 1 - 1.0545 \\ \rho_{BC} &= -0.0545 \qquad \Rightarrow \qquad \boxed{\rho_{BC} \approx -0.05}\end{aligned}$$

Conclusion

$$\rho_{AB} \approx -0.08, \qquad \rho_{AC} \approx 0.76, \qquad \rho_{BC} \approx -0.05$$

The highest (strongly positive) rank correlation is between A and C.

Judges A and C have the same liking in music.

2 Regression Lines

Regression coefficients from the raw sums

Regression of Y on X: $Y - \bar{Y} = b_{Y|X}(X - \bar{X})$, $\quad b_{Y|X} = \dfrac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$

Regression of X on Y: $X - \bar{X} = b_{X|Y}(Y - \bar{Y})$, $\quad b_{X|Y} = \dfrac{n\sum xy - (\sum x)(\sum y)}{n\sum y^2 - (\sum y)^2}$

$$\boxed{b_{Y|X}\, b_{X|Y} = r^2} \qquad \text{and} \qquad b_{Y|X} = r\frac{\sigma_Y}{\sigma_X}, \qquad b_{X|Y} = r\frac{\sigma_X}{\sigma_Y}$$

How to identify which line is which

Of the two given lines, the assignment of $b_{Y|X}$ and $b_{X|Y}$ must satisfy $b_{Y|X}\, b_{X|Y} = r^2 \leq 1$. If a choice gives $r^2 > 1$, the other assignment is the correct one. Both coefficients always carry the same sign, which is the sign of r.

2.1 Problem 1 – Engine age and maintenance cost

Problem 1

The data below show the **age of an engine** (in years) and the corresponding **maintenance expenditure** (in Rs. thousands) required to sustain its performance:

Engine Age x (years)	1	3	5	7	9
Annual Maintenance Cost y (Rs. $\times 1000$)	15	18	21	23	22

Form a **regression equation** between engine age and maintenance cost, and use it to **predict the maintenance cost** for a **4-year-old engine**.

Solution – Table of summations ($n = 5$)

x	y	x^2	xy
1	15	1	15
3	18	9	54
5	21	25	105
7	23	49	161
9	22	81	198
$\sum x = 25$	$\sum y = 99$	$\sum x^2 = 165$	$\sum xy = 533$

Means

$$\bar{x} = \frac{\sum x}{n} = \frac{25}{5}$$

$$\bar{x} = 5$$

$$\bar{y} = \frac{\sum y}{n} = \frac{99}{5}$$

$$\bar{y} = 19.8$$

Regression coefficient of y on x

$$b_{Y|X} = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

$$= \frac{5(533) - (25)(99)}{5(165) - (25)^2}$$

$$= \frac{2665 - 2475}{825 - 625}$$

$$= \frac{190}{200}$$

$$b_{Y|X} = 0.95$$

Regression equation of y on x

$$y - \bar{y} = b_{Y|X}\,(x - \bar{x})$$
$$y - 19.8 = 0.95\,(x - 5)$$
$$y - 19.8 = 0.95x - 4.75$$
$$y = 0.95x - 4.75 + 19.8$$
$$y = 0.95x + 15.05 \quad \Rightarrow \quad \boxed{y = 0.95x + 15.05}$$

Prediction for a 4-year-old engine ($x = 4$)

$$y = 0.95(4) + 15.05$$
$$= 3.80 + 15.05$$
$$y = 18.85 \quad \Rightarrow \quad \boxed{y = 18.85}$$

Conclusion

Regression equation: $y = 0.95x + 15.05$.

Predicted maintenance cost of a 4-year-old engine $= 18.85 \times 1000 =$ **Rs. 18,850**.

2.2 Problem 2 – Engine speed and fuel consumption

Problem 2

In an engine performance study, partial data were lost, but the following information about **engine speed** (X, in rpm) and **fuel consumption** (Y, in L/hr) was recovered:

i) $\mathrm{Var}(X) = 25$

ii) Regression equation of X on Y: $5x - y = 22$

iii) Regression equation of Y on X: $64x - 45y = 24$

Find:

1. The **mean engine speed** and **mean fuel consumption**.
2. The **correlation coefficient** between speed and fuel consumption.
3. The **standard deviation** of fuel consumption.

1. Mean engine speed and mean fuel consumption

Both regression lines pass through $(\bar{x}, \bar{y})$.

$$5\bar{x} - \bar{y} = 22 \qquad \cdots(1)$$

$$64\bar{x} - 45\bar{y} = 24 \qquad \cdots(2)$$

$$(1) \times 45: \quad 225\bar{x} - 45\bar{y} = 990 \qquad \cdots(3)$$

$$(3) - (2): \quad 161\bar{x} = 966$$

$$\bar{x} = \frac{966}{161} = 6 \quad \Rightarrow \quad \boxed{\bar{x} = 6}$$

$$\text{From (1)}: \quad \bar{y} = 5(6) - 22$$

$$\bar{y} = 8 \quad \Rightarrow \quad \boxed{\bar{y} = 8}$$

2. Correlation coefficient

$$X \text{ on } Y: \quad 5x - y = 22 \Rightarrow x = \frac{1}{5}y + \frac{22}{5}$$

$$b_{X|Y} = \frac{1}{5} = 0.2$$

$$Y \text{ on } X: \quad 64x - 45y = 24 \Rightarrow y = \frac{64}{45}x - \frac{24}{45}$$

$$b_{Y|X} = \frac{64}{45} = 1.4222$$

$$r = \sqrt{b_{X|Y}\, b_{Y|X}}$$

$$= \sqrt{\frac{1}{5} \times \frac{64}{45}}$$

$$= \sqrt{\frac{64}{225}}$$

$$= \frac{8}{15}$$

$$r = 0.5333 \quad \Rightarrow \quad \boxed{r \approx 0.53}$$

3. Standard deviation of fuel consumption

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{25} = 5$$

$$b_{Y|X} = r\,\frac{\sigma_Y}{\sigma_X}$$

$$\frac{64}{45} = \frac{8}{15} \times \frac{\sigma_Y}{5}$$

$$\sigma_Y = \frac{64}{45} \times 5 \times \frac{15}{8}$$

$$= \frac{4800}{360}$$

$$\sigma_Y = 13.33 \qquad \Rightarrow \qquad \boxed{\sigma_Y \approx 13.33}$$

Conclusion

$$\bar{x} = 6 \text{ rpm}, \quad \bar{y} = 8 \text{ L/hr}, \quad r = \tfrac{8}{15} \approx 0.53, \quad \sigma_Y \approx 13.33 \text{ L/hr}$$

2.3 Problem 3 – Heat transfer and flow rate

Problem 3

The two empirical correlations between **heat transfer** (Q) and **flow rate** (F) are given by:

$$8Q - 10F + 66 = 0 \qquad \text{and} \qquad 40Q - 18F - 214 = 0$$

Also, the **variance of heat transfer** is $\text{Var}(Q) = 9$.

Find:

i) The **mean values** of heat transfer and flow rate,

ii) The **coefficient of correlation** between heat transfer and flow rate, and

iii) The **standard deviation of flow rate**.

i) Mean values

Both lines pass through $(\bar{Q}, \bar{F})$.

$$8\bar{Q} - 10\bar{F} = -66 \qquad \cdots(1)$$

$$40\bar{Q} - 18\bar{F} = 214 \qquad \cdots(2)$$

$$(1) \times 5: \quad 40\bar{Q} - 50\bar{F} = -330 \qquad \cdots(3)$$

$$(2) - (3): \quad 32\bar{F} = 544$$

$$\bar{F} = \frac{544}{32} = 17 \quad \Rightarrow \quad \boxed{\bar{F} = 17}$$

$$\text{From (1)}: \quad 8\bar{Q} = 10(17) - 66 = 104$$

$$\bar{Q} = \frac{104}{8} = 13 \quad \Rightarrow \quad \boxed{\bar{Q} = 13}$$

ii) Coefficient of correlation

$$(1) \text{ as } F \text{ on } Q: \quad 10F = 8Q + 66 \Rightarrow F = 0.8\,Q + 6.6$$

$$b_{F|Q} = 0.8$$

$$(2) \text{ as } Q \text{ on } F: \quad 40Q = 18F + 214 \Rightarrow Q = 0.45\,F + 5.35$$

$$b_{Q|F} = 0.45$$

$$r = \sqrt{b_{F|Q}\, b_{Q|F}}$$

$$= \sqrt{0.8 \times 0.45}$$

$$= \sqrt{0.36} \quad (0.36 \leq 1)$$

$$r = 0.6 \quad \Rightarrow \quad \boxed{r = 0.6}$$

iii) Standard deviation of flow rate

$$\sigma_Q = \sqrt{\text{Var}(Q)} = \sqrt{9} = 3$$

$$b_{F|Q} = r\,\frac{\sigma_F}{\sigma_Q}$$

$$0.8 = 0.6 \times \frac{\sigma_F}{3}$$

$$\sigma_F = \frac{0.8 \times 3}{0.6}$$

$$= \frac{2.4}{0.6}$$

$$\sigma_F = 4 \quad \Rightarrow \quad \boxed{\sigma_F = 4}$$

Conclusion

$$\bar{Q} = 13, \quad \bar{F} = 17, \quad r = 0.6, \quad \sigma_F = 4$$

3 Statistical Quality Control – Overview

What is SQC?

Statistical Quality Control (SQC) uses samples taken from a process to decide whether the variation in the process is due only to **chance causes** (process *in control*) or also to **assignable causes** (process *out of control*). Its main tool is the **control chart**: a sample statistic plotted against the sample number together with a centre line (CL) and an upper and a lower control limit (UCL, LCL).

General rule (3-sigma limits): $CL = \mu_s, \quad UCL = \mu_s + 3\sigma_s, \quad LCL = \mu_s - 3\sigma_s,$

where μ_s and σ_s are the mean and standard error of the plotted statistic.

The five control charts at a glance

Chart	Data	Plotted for sample i	CL	$UCL,\ LCL$
$\bar{X}$	Variable (measured)	mean $\bar{X}_i$	$\bar{\bar{X}}$	$\bar{\bar{X}} \pm A_2\bar{R}$
R	Variable (measured)	range R_i	$\bar{R}$	$D_4\bar{R},\ D_3\bar{R}$
p	Attribute: defectives	proportion $p_i = d_i/n$	$\bar{p}$	$\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$
np	Attribute: defectives	number d_i	$n\bar{p}$	$n\bar{p} \pm 3\sqrt{n\bar{p}(1-\bar{p})}$
c	Attribute: defects	count c_i	$\bar{c}$	$\bar{c} \pm 3\sqrt{\bar{c}}$

A *defective* is a unit that fails the specification (p, np charts); a *defect* is a single flaw, and one unit may carry several (c-chart). A negative LCL is always taken as 0.

Reading a control chart

- **In control:** every point lies between LCL and UCL and the points show no systematic pattern – only chance causes are present.
- **Out of control:** at least one point lies outside the limits (or a clear pattern appears, such as a long run on one side of CL or a steady trend) – look for an assignable cause at that sample.

Control chart constants used in this unit

Sample size n	A_2	D_3	D_4
3	1.023	0	2.574
5	0.577	0	2.114
6	0.483	0	2.004

4 $\bar{X}$-Chart (Mean Chart)

Overview

- **Purpose:** monitors the *process average* (central tendency).
- **Data:** k samples, each of n measurements $(X_{i1}, \ldots, X_{in})$.
- **Plotted:** the sample means $\bar{X}_i$ against the sample number.
- **Companion:** always read together with the R-chart, which checks the spread.

Formulae for the $\bar{X}$-chart

$$\bar{X}_i = \frac{\sum_{j=1}^{n} X_{ij}}{n}, \qquad R_i = X_{\max} - X_{\min},$$

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k} \bar{X}_i}{k}, \qquad \bar{R} = \frac{\sum_{i=1}^{k} R_i}{k},$$

$$UCL = \bar{\bar{X}} + A_2\bar{R}, \qquad CL = \bar{\bar{X}}, \qquad LCL = \bar{\bar{X}} - A_2\bar{R}.$$

A_2 is read from the table of constants for the sample size n.

4.1 Problem 1 – Diameter of bearings

Problem 1

A factory produces bearings, and the diameter of the bearings is critical to their performance. The factory collects data on the diameter of 5 bearings every hour for 10 hours. The data is shown below:

Sample	Diameter (mm)
1	10.01, 10.03, 9.99, 10.00, 10.02
2	10.05, 10.06, 10.04, 10.03, 10.01
3	9.98, 9.97, 9.99, 10.01, 10.02
4	10.00, 10.01, 9.99, 10.00, 10.03
5	10.02, 10.00, 10.01, 9.99, 10.00
6	9.99, 10.00, 9.98, 10.02, 9.99
7	10.01, 10.03, 10.00, 10.02, 9.99
8	10.03, 10.04, 10.02, 10.00, 10.01
9	10.00, 9.99, 10.01, 10.02, 9.98
10	9.97, 9.98, 10.00, 10.01, 10.02

Check whether the given process is under control.

Step 1: Mean and range of each sample (sample 1 shown in full)

$$\bar{X}_i = \frac{\sum_{j=1}^{n} X_{ij}}{n}$$

$$\bar{X}_1 = \frac{10.01 + 10.03 + 9.99 + 10.00 + 10.02}{5}$$

$$= \frac{50.05}{5}$$

$$= 10.010$$

$$R_i = X_{\max} - X_{\min}$$

$$R_1 = 10.03 - 9.99$$

$$= 0.04$$

Step 1 (contd.): Table of $\bar{X}_i$ and R_i

Sample i	$\sum X$	$\bar{X}_i = \dfrac{\sum X}{5}$	$X_{\max}$	$X_{\min}$	$R_i = X_{\max} - X_{\min}$
1	50.05	10.010	10.03	9.99	0.04
2	50.19	10.038	10.06	10.01	0.05
3	49.97	9.994	10.02	9.97	0.05
4	50.03	10.006	10.03	9.99	0.04
5	50.02	10.004	10.02	9.99	0.03
6	49.98	9.996	10.02	9.98	0.04
7	50.05	10.010	10.03	9.99	0.04
8	50.10	10.020	10.04	10.00	0.04
9	50.00	10.000	10.02	9.98	0.04
10	49.98	9.996	10.02	9.97	0.05
	Total	100.074			0.42

Step 2: Grand mean $\bar{\bar{X}}$ and average range $\bar{R}$ ($k = 10$ samples)

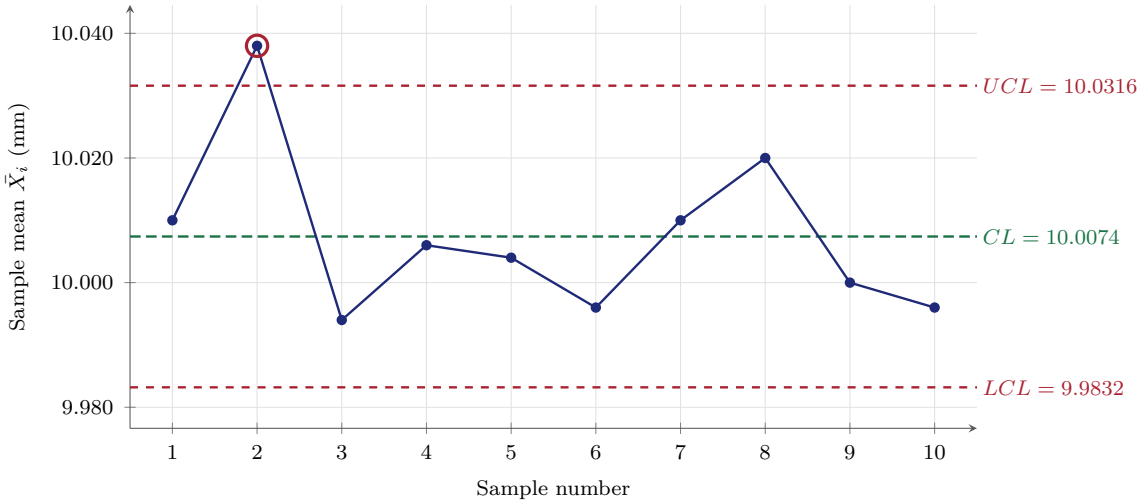
$$
\begin{aligned}
\bar{\bar{X}} &= \frac{\sum_{i=1}^{k} \bar{X}_i}{k} \\
&= \frac{1}{10}\big(10.010 + 10.038 + 9.994 + 10.006 + 10.004 \\
&\qquad + 9.996 + 10.010 + 10.020 + 10.000 + 9.996\big) \\
&= \frac{100.074}{10} \\
&= \boxed{10.0074}
\end{aligned}
$$

$$
\begin{aligned}
\bar{R} &= \frac{\sum_{i=1}^{k} R_i}{k} \\
&= \frac{1}{10}\big(0.04 + 0.05 + 0.05 + 0.04 + 0.03 + 0.04 \\
&\qquad + 0.04 + 0.04 + 0.04 + 0.05\big) \\
&= \frac{0.42}{10} \\
&= \boxed{0.042}
\end{aligned}
$$

Step 3: Control limits of the $\bar{X}$-chart ($n = 5$: $A_2 = 0.577$)

$$\begin{aligned} A_2\bar{R} &= (0.577)(0.042) \\ &= 0.0242 \\ UCL &= \bar{\bar{X}} + A_2\bar{R} \\ &= 10.0074 + 0.0242 \\ &= \boxed{10.0316} \\ CL &= \bar{\bar{X}} \\ &= \boxed{10.0074} \\ LCL &= \bar{\bar{X}} - A_2\bar{R} \\ &= 10.0074 - 0.0242 \\ &= \boxed{9.9832} \end{aligned}$$

Step 4: Control chart (Sample mean $\bar{X}_i$ plotted against sample number)



Supporting check on variability (R-chart limits)

$$\begin{aligned} UCL_R &= D_4\bar{R} \\ &= (2.114)(0.042) \\ &= 0.0888 \\ LCL_R &= D_3\bar{R} = 0 \end{aligned}$$

Largest range $= 0.05 < 0.0888$, so every R_i lies within the R-chart limits: **process variability is in control**.

Conclusion – process NOT in control

Sample 2 has $\bar{X}_2 = 10.038$, which lies **above the upper control limit** ($UCL = 10.0316$, circled on the chart). The other 9 sample means lie between 9.9832 and 10.0316, and all ranges are within their limits.

The process average is out of control; sample 2 must be investigated for an assignable cause.

4.2 Problem 2 – Thickness of metal sheets

Problem 2

A manufacturer monitors the average thickness of metal sheets produced in their factory. Every 30 minutes, they measure the thickness of 6 metal sheets. The data below shows the measured thickness (in mm) over 8 samples:

Sample	Thickness (mm)
1	1.02, 1.00, 1.01, 1.03, 1.01, 1.02
2	1.04, 1.02, 1.01, 1.03, 1.02, 1.05
3	1.01, 1.00, 1.03, 1.02, 1.04, 1.01
4	1.02, 1.03, 1.01, 1.04, 1.00, 1.02
5	1.03, 1.01, 1.02, 1.03, 1.05, 1.04
6	1.02, 1.00, 1.01, 1.03, 1.04, 1.05
7	1.01, 1.03, 1.00, 1.02, 1.01, 1.04
8	1.04, 1.02, 1.01, 1.05, 1.03, 1.02

Check whether the given process is under control.

Step 1: Mean and range of each sample (sample 1 shown in full)

$$\begin{aligned}
\bar{X}_i &= \frac{\sum_{j=1}^{n} X_{ij}}{n} \\
\bar{X}_1 &= \frac{1.02 + 1.00 + 1.01 + 1.03 + 1.01 + 1.02}{6} \\
&= \frac{6.09}{6} \\
&= 1.0150 \\
R_i &= X_{\max} - X_{\min} \\
R_1 &= 1.03 - 1.00 \\
&= 0.03
\end{aligned}$$

Step 1 (contd.): Table of $\bar{X}_i$ and R_i

Sample i	$\sum X$	$\bar{X}_i = \dfrac{\sum X}{6}$	$X_{\max}$	$X_{\min}$	$R_i = X_{\max} - X_{\min}$
1	6.09	1.0150	1.03	1.00	0.03
2	6.17	1.0283	1.05	1.01	0.04
3	6.11	1.0183	1.04	1.00	0.04
4	6.12	1.0200	1.04	1.00	0.04
5	6.18	1.0300	1.05	1.01	0.04
6	6.15	1.0250	1.05	1.00	0.05
7	6.11	1.0183	1.04	1.00	0.04
8	6.17	1.0283	1.05	1.01	0.04
	Total	8.1832			0.32

Step 2: Grand mean $\bar{\bar{X}}$ and average range $\bar{R}$ ($k = 8$ samples)

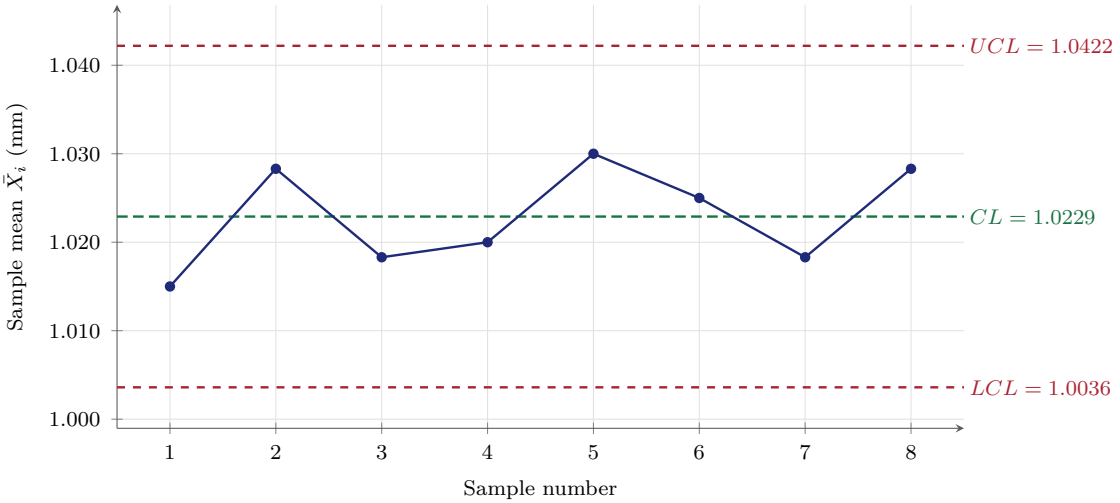
$$
\begin{aligned}
\bar{\bar{X}} &= \frac{\sum_{i=1}^{k} \bar{X}_i}{k} \\
&= \frac{1}{8}\Big(1.0150 + 1.0283 + 1.0183 + 1.0200 + 1.0300 \\
&\qquad + 1.0250 + 1.0183 + 1.0283\Big) \\
&= \frac{8.1832}{8} \\
&= \boxed{1.0229}
\end{aligned}
$$

$$
\begin{aligned}
\bar{R} &= \frac{\sum_{i=1}^{k} R_i}{k} \\
&= \frac{1}{8}\Big(0.03 + 0.04 + 0.04 + 0.04 + 0.04 + 0.05 \\
&\qquad + 0.04 + 0.04\Big) \\
&= \frac{0.32}{8} \\
&= \boxed{0.040}
\end{aligned}
$$

Step 3: Control limits of the $\bar{X}$-chart ($n = 6$: $A_2 = 0.483$)

$$\begin{aligned} A_2\bar{R} &= (0.483)(0.040) \\ &= 0.0193 \\ UCL &= \bar{\bar{X}} + A_2\bar{R} \\ &= 1.0229 + 0.0193 \\ &= \boxed{1.0422} \\ CL &= \bar{\bar{X}} \\ &= \boxed{1.0229} \\ LCL &= \bar{\bar{X}} - A_2\bar{R} \\ &= 1.0229 - 0.0193 \\ &= \boxed{1.0036} \end{aligned}$$

Step 4: Control chart (Sample mean $\bar{X}_i$ plotted against sample number)



Supporting check on variability (R-chart limits)

$$\begin{aligned} UCL_R &= D_4\bar{R} \\ &= (2.004)(0.040) \\ &= 0.0802 \\ LCL_R &= D_3\bar{R} = 0 \end{aligned}$$

Largest range $= 0.05 < 0.0802$, so every R_i lies within the R-chart limits: **process variability is in control**.

Conclusion – process in control

All 8 sample means lie between $LCL = 1.0036$ and $UCL = 1.0422$ (smallest 1.0150, largest 1.0300) with no trend, and all ranges are within the R-chart limits.

The process is in control.

5 R-Chart (Range Chart)

Overview

- **Purpose:** monitors the *process variability* (spread).
- **Data:** k samples, each of n measurements.
- **Plotted:** the sample ranges $R_i = X_{\max} - X_{\min}$ against the sample number.
- **Note:** a point above UCL signals increased variability; for $n \le 6$, $D_3 = 0$, so $LCL = 0$.

Formulae for the R-chart

$$R_i = X_{\max} - X_{\min}, \qquad \bar{R} = \frac{\sum_{i=1}^{k} R_i}{k},$$

$$UCL = D_4\bar{R}, \qquad CL = \bar{R}, \qquad LCL = D_3\bar{R}.$$

D_3, D_4 are read from the table of constants for the sample size n.

5.1 Problem 1 – Construction of an R-Chart for a Manufacturing Process

Problem 1

Problem Statement: A factory is monitoring the consistency of the diameter of steel rods. Five samples are taken every hour, and the results from six consecutive hours are as follows (diameter in millimeters):

Hour	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
1	4.90	5.10	5.00	5.05	5.02
2	4.92	5.08	5.01	5.04	5.00
3	5.01	5.04	5.02	5.03	5.00
4	4.98	5.06	5.01	5.02	5.03
5	5.02	5.07	5.03	5.00	5.01
6	5.00	5.02	5.03	5.05	5.01

Check the variability of the process.

Step 1: Mean and range of each sample (sample 1 shown in full)

$$\bar{X}_i = \frac{\sum_{j=1}^{n} X_{ij}}{n}$$

$$\bar{X}_1 = \frac{4.90 + 5.10 + 5.00 + 5.05 + 5.02}{5}$$

$$= \frac{25.07}{5}$$

$$= 5.014$$

$$R_i = X_{\max} - X_{\min}$$

$$R_1 = 5.10 - 4.90$$

$$= 0.20$$

Step 1 (contd.): Table of $\bar{X}_i$ and R_i

Hour i	$\sum X$	$\bar{X}_i = \frac{\sum X}{5}$	$X_{\max}$	$X_{\min}$	$R_i = X_{\max} - X_{\min}$
1	25.07	5.014	5.10	4.90	0.20
2	25.05	5.010	5.08	4.92	0.16
3	25.10	5.020	5.04	5.00	0.04
4	25.10	5.020	5.06	4.98	0.08
5	25.13	5.026	5.07	5.00	0.07
6	25.11	5.022	5.05	5.00	0.05
	Total	30.112			0.60

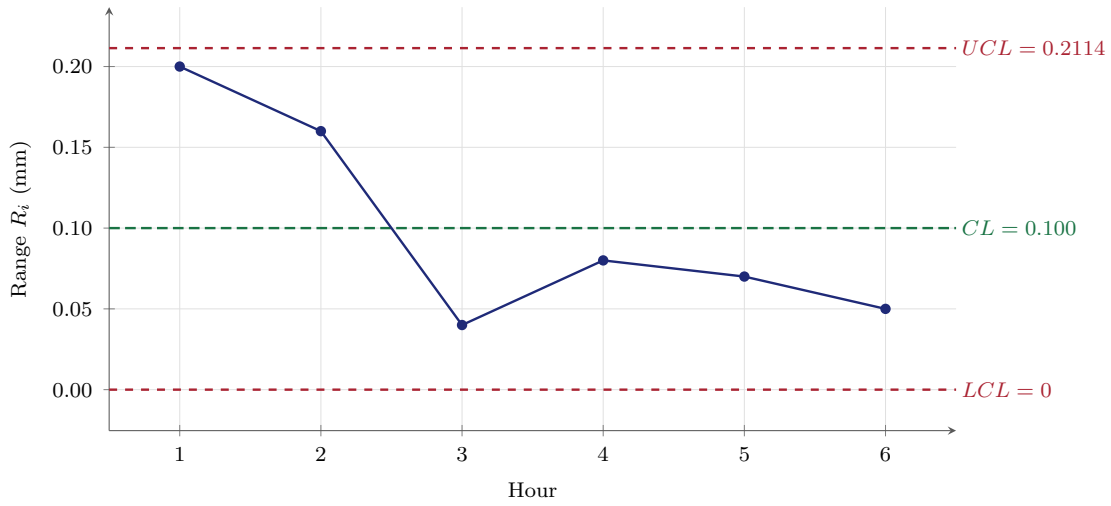
Step 2: Grand mean $\bar{\bar{X}}$ and average range $\bar{R}$ ($k = 6$ samples)

$$\begin{aligned}
\bar{\bar{X}} &= \frac{\sum_{i=1}^{k} \bar{X}_i}{k} \\
&= \frac{1}{6}\big(5.014 + 5.010 + 5.020 + 5.020 + 5.026 \\
&\qquad + 5.022\big) \\
&= \frac{30.112}{6} \\
&= \boxed{5.0187} \\
\bar{R} &= \frac{\sum_{i=1}^{k} R_i}{k} \\
&= \frac{0.20 + 0.16 + 0.04 + 0.08 + 0.07 + 0.05}{6} \\
&= \frac{0.60}{6} \\
&= \boxed{0.100}
\end{aligned}$$

Step 3: Control limits of the R-chart ($n = 5$: $D_3 = 0$, $D_4 = 2.114$)

$$\begin{aligned}
UCL &= D_4 \bar{R} \\
&= (2.114)(0.100) \\
&= \boxed{0.2114} \\
CL &= \bar{R} \\
&= \boxed{0.100} \\
LCL &= D_3 \bar{R} \\
&= (0)(0.100) \\
&= \boxed{0}
\end{aligned}$$

Step 4: Control chart (Range R_i plotted against hour)



Conclusion – variability in control

All 6 ranges lie between $LCL = 0$ and $UCL = 0.2114$; the largest is $R_1 = 0.20$.

The process variability is in control (stable).

5.2 Problem 2 – R-Chart for Monitoring Temperature Variation in a Chemical Process

Problem 2

Problem Statement: In a chemical process, the temperature is monitored to maintain process stability. Three samples are taken at four different time intervals, and the temperatures are recorded as follows (in degrees Celsius):

Time Interval	Sample 1	Sample 2	Sample 3
1	78.0	80.5	79.5
2	79.0	79.2	80.0
3	78.5	79.8	79.3
4	79.1	79.7	78.8

Check the variability of the process.

Step 1: Mean and range of each sample (sample 1 shown in full)

$$\bar{X}_i = \frac{\sum_{j=1}^{n} X_{ij}}{n}$$

$$\bar{X}_1 = \frac{78.0 + 80.5 + 79.5}{3}$$

$$= \frac{238.0}{3}$$

$$= 79.333$$

$$R_i = X_{\max} - X_{\min}$$

$$R_1 = 80.5 - 78.0$$

$$= 2.5$$

Step 1 (contd.): Table of $\bar{X}_i$ and R_i

Time Interval i	$\sum X$	$\bar{X}_i = \frac{\sum X}{3}$	$X_{\max}$	$X_{\min}$	$R_i = X_{\max} - X_{\min}$
1	238.0	79.333	80.5	78.0	2.5
2	238.2	79.400	80.0	79.0	1.0
3	237.6	79.200	79.8	78.5	1.3
4	237.6	79.200	79.7	78.8	0.9
	Total	317.133			5.7

Step 2: Grand mean $\bar{\bar{X}}$ and average range $\bar{R}$ ($k = 4$ samples)

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k} \bar{X}_i}{k}$$

$$= \frac{79.333 + 79.400 + 79.200 + 79.200}{4}$$

$$= \frac{317.133}{4}$$

$$= \boxed{79.2832}$$

$$\bar{R} = \frac{\sum_{i=1}^{k} R_i}{k}$$

$$= \frac{2.5 + 1.0 + 1.3 + 0.9}{4}$$

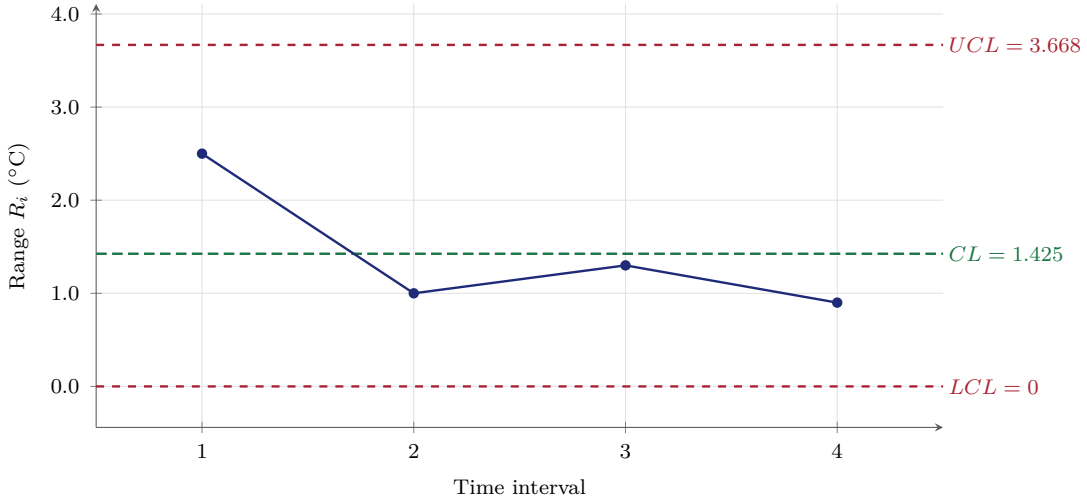
$$= \frac{5.7}{4}$$

$$= \boxed{1.425}$$

Step 3: Control limits of the R-chart ($n = 3$: $D_3 = 0$, $D_4 = 2.574$)

$$\begin{aligned} UCL &= D_4 \bar{R} \\ &= (2.574)(1.425) \\ &= \boxed{3.668} \\ CL &= \bar{R} \\ &= \boxed{1.425} \\ LCL &= D_3 \bar{R} \\ &= (0)(1.425) \\ &= \boxed{0} \end{aligned}$$

Step 4: Control chart (Range R_i plotted against time interval)



Conclusion – variability in control

All 4 ranges lie between $LCL = 0$ and $UCL = 3.668$; the largest is $R_1 = 2.5$.

The process variability is in control (stable).

6 p-Chart (Proportion Defective)

Overview

- **Purpose:** monitors the *proportion of defective units.*
- **Data:** attribute data – number of defectives d_i in each of k samples of size n.
- **Plotted:** $p_i = d_i/n$ against the sample number.

Formulae for the p-chart

$$p_i = \frac{d_i}{n}, \qquad \bar{p} = \frac{\sum d_i}{n\,k} = \frac{\text{total defectives}}{\text{total units inspected}},$$

$$\sigma_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}},$$

$$UCL = \bar{p} + 3\sigma_p, \qquad CL = \bar{p}, \qquad LCL = \bar{p} - 3\sigma_p.$$

If LCL comes out negative it is taken as 0.

6.1 Problem 1 – Defective units in batches of 200

Problem 1

A factory produces batches of 200 units each. The management wants to monitor the proportion of defective units over 10 consecutive days. The number of defective units per batch for each day is recorded as follows:

Day	1	2	3	4	5	6	7	8	9	10
Defectives	6	4	5	9	3	5	7	6	4	8

We need to create a p-chart to monitor the proportion of defective units and determine if the process is under control.

Step 1: Proportion defective of each sample

$$p_i = \frac{d_i}{n}$$

$$p_1 = \frac{6}{200} = 0.030$$

Day	1	2	3	4	5	6	7	8	9	10
p_i	0.030	0.020	0.025	0.045	0.015	0.025	0.035	0.030	0.020	0.040

Step 2: Average proportion defective $\bar{p}$

$$\bar{p} = \frac{\sum d_i}{n\,k} = \frac{6+4+5+9+3+5+7+6+4+8}{200 \times 10} = \frac{57}{2000} = \boxed{0.0285}$$

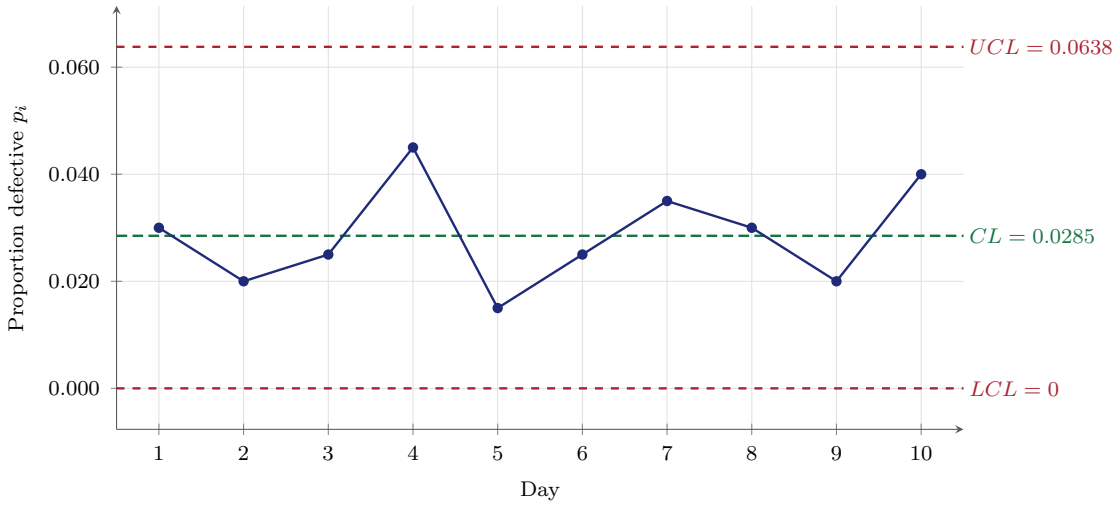
Step 3: Standard error σ_p $(n = 200)$

$$\begin{aligned}\sigma_p &= \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \\ &= \sqrt{\frac{(0.0285)(0.9715)}{200}} \\ &= \sqrt{\frac{0.02768775}{200}} \\ &= \sqrt{0.00013844} \\ &= 0.011766\end{aligned}$$

Step 4: Control limits

$$\begin{aligned}UCL &= \bar{p} + 3\sigma_p \\ &= 0.0285 + 3(0.011766) \\ &= 0.0285 + 0.0353 \\ &= \boxed{0.0638} \\ CL &= \bar{p} \\ &= \boxed{0.0285} \\ LCL &= \bar{p} - 3\sigma_p \\ &= 0.0285 - 0.0353 \\ &= -0.0068 \\ &< 0 \\ \therefore\ LCL &= \boxed{0}\end{aligned}$$

Step 5: Control chart



Conclusion – process in control

All 10 points lie between $LCL = 0$ and $UCL = 0.0638$; the highest value is 0.045 (day 4).

The process is in control – only chance causes of variation are present.

6.2 Problem 2 – Errors in insurance claims

Problem 2

An insurance company reviews 100 claims per day over 12 days to monitor errors in processing. The number of claims with errors per day is shown below:

Day	1	2	3	4	5	6	7	8	9	10	11	12
Errors	3	1	2	4	3	2	5	3	1	4	3	2

We will construct a p-chart to monitor the error rate and check if the process is in control.

Step 1: Proportion defective of each sample

$$p_i = \frac{d_i}{n}$$

$$p_1 = \frac{3}{100} = 0.030$$

Day	1	2	3	4	5	6	7	8	9	10	11	12
p_i	0.030	0.010	0.020	0.040	0.030	0.020	0.050	0.030	0.010	0.040	0.030	0.020

Step 2: Average proportion defective $\bar{p}$

$$\bar{p} = \frac{\sum d_i}{n\,k} = \frac{3+1+2+4+3+2+5+3+1+4+3+2}{100 \times 12} = \frac{33}{1200} = \boxed{0.0275}$$

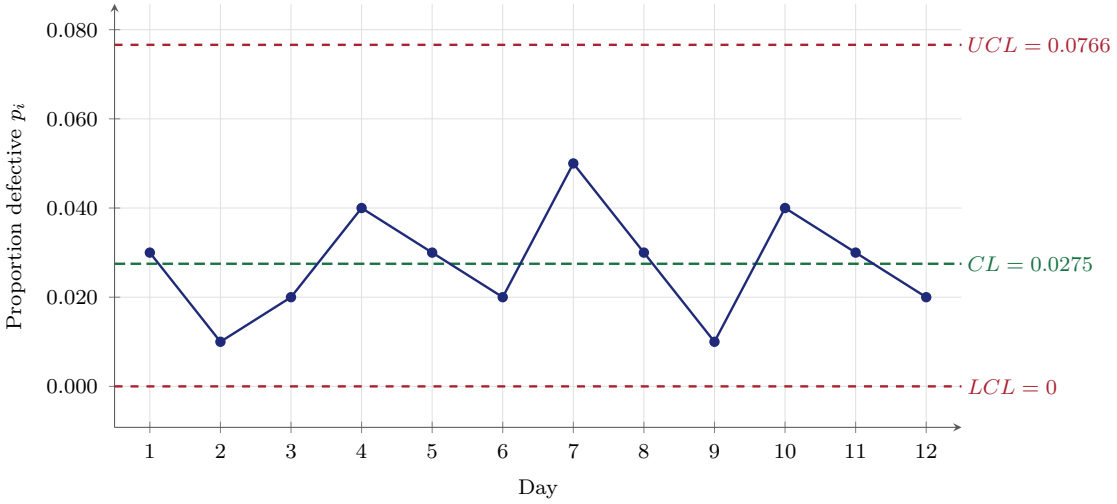
Step 3: Standard error σ_p $(n = 100)$

$$\begin{aligned}\sigma_p &= \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \\ &= \sqrt{\frac{(0.0275)(0.9725)}{100}} \\ &= \sqrt{\frac{0.02674375}{100}} \\ &= \sqrt{0.00026744} \\ &= 0.016354\end{aligned}$$

Step 4: Control limits

$$\begin{aligned}UCL &= \bar{p} + 3\sigma_p \\ &= 0.0275 + 3(0.016354) \\ &= 0.0275 + 0.0491 \\ &= \boxed{0.0766} \\ CL &= \bar{p} \\ &= \boxed{0.0275} \\ LCL &= \bar{p} - 3\sigma_p \\ &= 0.0275 - 0.0491 \\ &= -0.0216 \\ &< 0 \\ \therefore\ LCL &= \boxed{0}\end{aligned}$$

Step 5: Control chart



Conclusion – process in control

All 12 points lie between $LCL = 0$ and $UCL = 0.0766$; the highest value is 0.050 (day 7).

The process is in control – only chance causes of variation are present.

7 np-Chart (Number of Defectives)

Overview

- **Purpose:** monitors the *number of defective units* in a sample.
- **Data:** number of defectives d_i in each of k samples of the *same* size n.
- **Plotted:** d_i itself against the sample number (same information as the p-chart, in counts).

Formulae for the np-chart

$$\bar{p} = \frac{\sum d_i}{n\,k}, \qquad CL = n\bar{p} = \frac{\sum d_i}{k},$$
$$\sigma_{np} = \sqrt{n\bar{p}\,(1-\bar{p})},$$
$$UCL = n\bar{p} + 3\sigma_{np}, \qquad LCL = n\bar{p} - 3\sigma_{np}.$$

If LCL comes out negative it is taken as 0.

7.1 Problem 1 – Defective light bulbs

Problem 1

A factory produces light bulbs in batches of 150. The number of defective bulbs per batch is recorded over 8 days:

Day	1	2	3	4	5	6	7	8
Defectives	7	5	6	10	4	8	6	9

We will create an np-chart to monitor the number of defective bulbs per batch.

Step 1: $\bar{p}$ and the centre line $n\bar{p}$

$$\begin{aligned}
\bar{p} &= \frac{\sum d_i}{n\,k} \\
&= \frac{7+5+6+10+4+8+6+9}{150 \times 8} \\
&= \frac{55}{1200} \\
&= 0.045833 \\
n\bar{p} &= n \times \bar{p} \\
&= 150 \times \frac{55}{1200} \\
&= \frac{55}{8} \\
&= \boxed{6.8750}
\end{aligned}$$

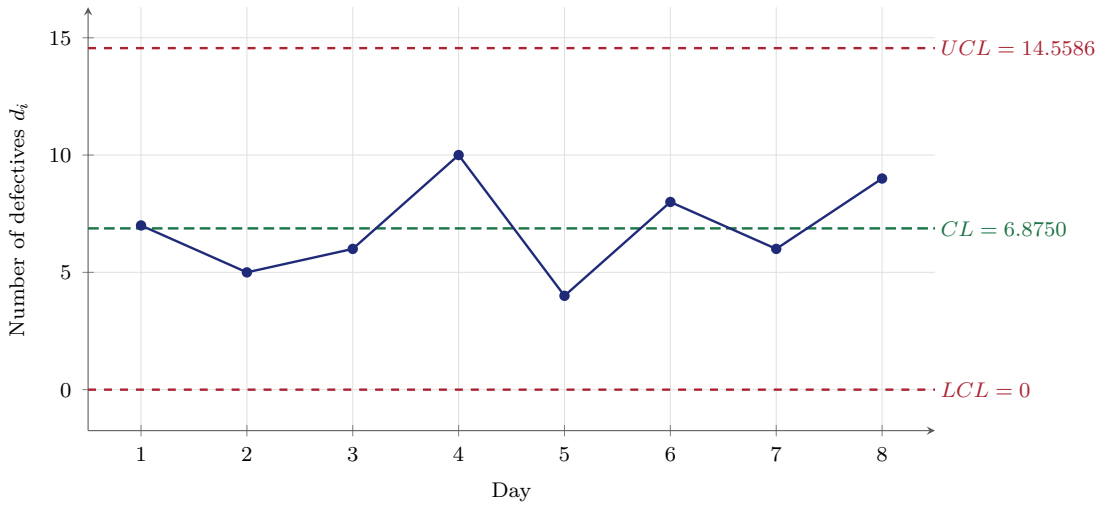
Step 2: Standard error σ_{np} $(n = 150)$

$$\begin{aligned}
\sigma_{np} &= \sqrt{n\bar{p}\,(1-\bar{p})} \\
&= \sqrt{(6.8750)(0.954167)} \\
&= \sqrt{6.5599} \\
&= 2.5612
\end{aligned}$$

Step 3: Control limits

$$\begin{aligned}
UCL &= n\bar{p} + 3\sigma_{np} \\
&= 6.8750 + 3(2.5612) \\
&= 6.8750 + 7.6836 \\
&= \boxed{14.5586} \\
CL &= n\bar{p} \\
&= \boxed{6.8750} \\
LCL &= n\bar{p} - 3\sigma_{np} \\
&= 6.8750 - 7.6836 \\
&= -0.8086 \\
&< 0 \\
\therefore\ LCL &= \boxed{0}
\end{aligned}$$

Step 4: Control chart



Conclusion – process in control

All 8 points lie between $LCL = 0$ and $UCL = 14.5586$; the highest value is 10 (day 4).

The process is in control – only chance causes of variation are present.

7.2 Problem 2 – Printing errors

Problem 2

Problem Description: A printing press examines 200 pages daily for printing errors. The number of pages with errors over 6 days is recorded below:

Day	1	2	3	4	5	6
Errors	5	3	4	8	2	6

We will construct an np-chart to monitor the number of defective pages.

Step 1: $\bar{p}$ and the centre line $n\bar{p}$

$$\begin{aligned}
\bar{p} &= \frac{\sum d_i}{n\,k} \\
&= \frac{5+3+4+8+2+6}{200 \times 6} \\
&= \frac{28}{1200} \\
&= 0.023333 \\
n\bar{p} &= n \times \bar{p} \\
&= 200 \times \frac{28}{1200} \\
&= \frac{28}{6} \\
&= \boxed{4.6667}
\end{aligned}$$

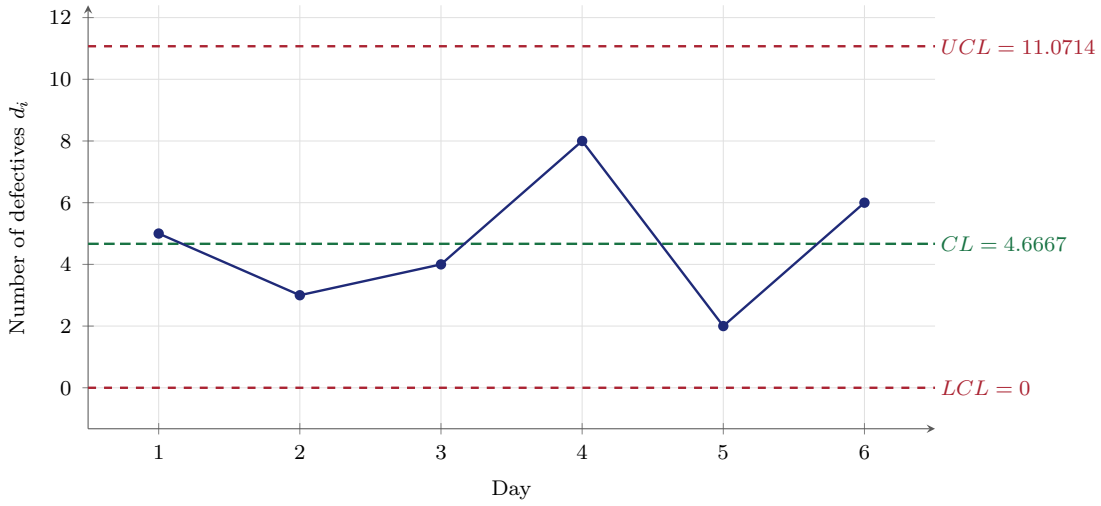
Step 2: Standard error σ_{np} $(n = 200)$

$$\begin{aligned}\sigma_{np} &= \sqrt{n\bar{p}\,(1-\bar{p})} \\ &= \sqrt{(4.6667)(0.976667)} \\ &= \sqrt{4.5578} \\ &= 2.1349\end{aligned}$$

Step 3: Control limits

$$\begin{aligned}UCL &= n\bar{p} + 3\sigma_{np} \\ &= 4.6667 + 3(2.1349) \\ &= 4.6667 + 6.4047 \\ &= \boxed{11.0714} \\ CL &= n\bar{p} \\ &= \boxed{4.6667} \\ LCL &= n\bar{p} - 3\sigma_{np} \\ &= 4.6667 - 6.4047 \\ &= -1.7380 \\ &< 0 \\ \therefore\ LCL &= \boxed{0}\end{aligned}$$

Step 4: Control chart



Conclusion – process in control

All 6 points lie between $LCL = 0$ and $UCL = 11.0714$; the highest value is 8 (day 4).

The process is in control – only chance causes of variation are present.

8 c-Chart (Number of Defects)

Overview

- **Purpose:** monitors the *number of defects* (nonconformities) per inspection unit.
- **Data:** defect count c_i in each of k inspection units of the same size; one unit may have many defects.
- **Basis:** counts follow a Poisson law, whose mean and variance are both $\bar{c}$.

Formulae for the c-chart

$$\bar{c} = \frac{\sum c_i}{k}, \qquad \sigma_c = \sqrt{\bar{c}},$$
$$UCL = \bar{c} + 3\sqrt{\bar{c}}, \qquad CL = \bar{c}, \qquad LCL = \bar{c} - 3\sqrt{\bar{c}}.$$

If LCL comes out negative it is taken as 0.

8.1 Problem 1 – Defects in Car Seats Production

Problem 1

A company manufactures car seats, and quality control is done by inspecting 100 car seats every day for defects. The defects observed in the last 10 days are as follows:

Day	1	2	3	4	5	6	7	8	9	10
Defects	5	3	6	4	7	3	4	8	5	4

Are the defects within acceptable control limits?

Step 1: Average number of defects $\bar{c}$ ($k = 10$)

$$\begin{aligned}\bar{c} &= \frac{\sum c_i}{k} \\ &= \frac{5+3+6+4+7+3+4+8+5+4}{10} \\ &= \frac{49}{10} \\ &= \boxed{4.9}\end{aligned}$$

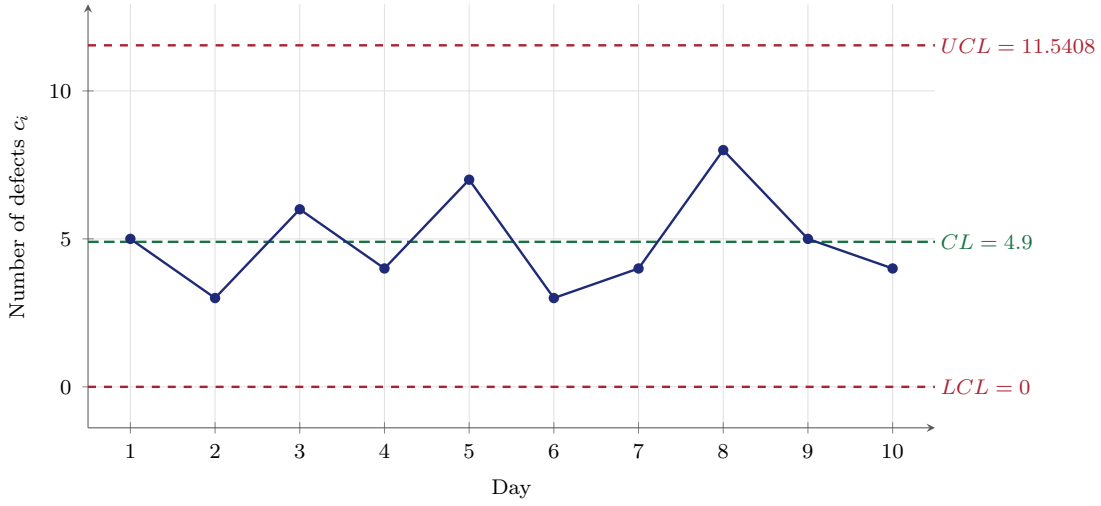
Step 2: Standard error $\sqrt{\bar{c}}$

$$\begin{aligned}\sigma_c &= \sqrt{\bar{c}} \\ &= \sqrt{4.9} \\ &= 2.2136\end{aligned}$$

Step 3: Control limits

$$
\begin{aligned}
UCL &= \bar{c} + 3\sqrt{\bar{c}} \\
&= 4.9 + 3(2.2136) \\
&= 4.9 + 6.6408 \\
&= \boxed{11.5408} \\
CL &= \bar{c} \\
&= \boxed{4.9} \\
LCL &= \bar{c} - 3\sqrt{\bar{c}} \\
&= 4.9 - 6.6408 \\
&= -1.7408 \\
&< 0 \\
\therefore\ LCL &= \boxed{0}
\end{aligned}
$$

Step 4: Control chart



Conclusion – process in control

All 10 points lie between $LCL = 0$ and $UCL = 11.5408$; the highest value is 8 (day 8). The counts fluctuate randomly about $CL = 4.9$ (5 above and 5 below), with no trend.

The process is in control – only chance causes of variation are present.

8.2 Problem 2 – Defects in a Printing Process

Problem 2

A printing company inspects sheets of printed material for defects. In each inspection, 50 sheets are examined. The number of defects observed over 12 consecutive inspections is as follows:

Inspection	1	2	3	4	5	6	7	8	9	10	11	12
Defects	2	1	3	2	4	2	3	5	4	3	6	2

Are the defects within acceptable control limits?

Step 1: Average number of defects $\bar{c}$ $(k = 12)$

$$\begin{aligned}\bar{c} &= \frac{\sum c_i}{k} \\ &= \frac{2+1+3+2+4+2+3+5+4+3+6+2}{12} \\ &= \frac{37}{12} \\ &= \boxed{3.0833}\end{aligned}$$

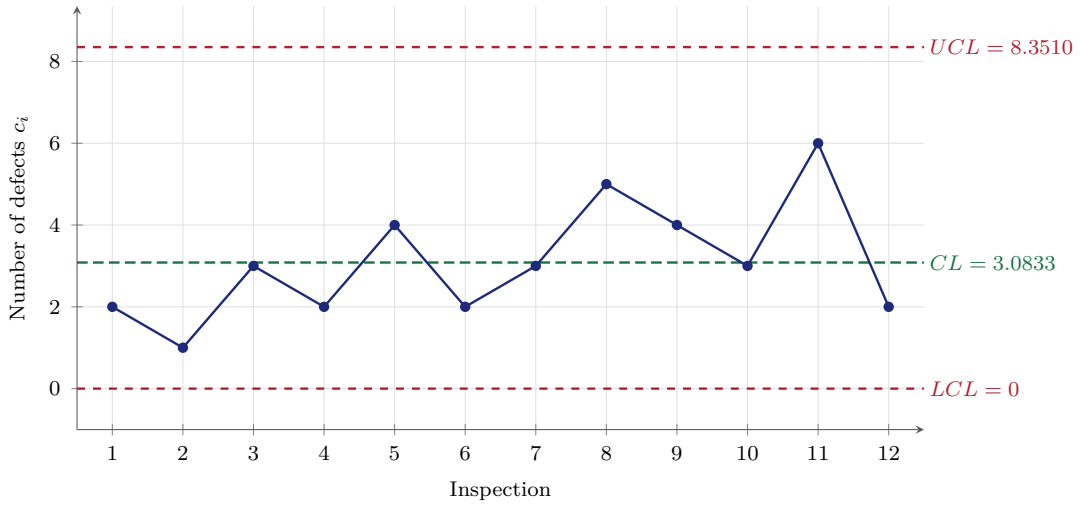
Step 2: Standard error $\sqrt{\bar{c}}$

$$\begin{aligned}\sigma_c &= \sqrt{\bar{c}} \\ &= \sqrt{3.0833} \\ &= 1.7559\end{aligned}$$

Step 3: Control limits

$$\begin{aligned}UCL &= \bar{c} + 3\sqrt{\bar{c}} \\ &= 3.0833 + 3(1.7559) \\ &= 3.0833 + 5.2677 \\ &= \boxed{8.3510} \\ CL &= \bar{c} \\ &= \boxed{3.0833} \\ LCL &= \bar{c} - 3\sqrt{\bar{c}} \\ &= 3.0833 - 5.2677 \\ &= -2.1844 \\ &< 0 \\ \therefore\ LCL &= \boxed{0}\end{aligned}$$

Step 4: Control chart



Conclusion – process in control

All 12 points lie between $LCL = 0$ and $UCL = 8.3510$; the highest value is 6 (inspection 11).

The process is in control – only chance causes of variation are present.

8.3 Exercise – Monitoring Defects in Bottling Process

Exercise

A beverage company is monitoring the defects in its bottling process. Every day, the company inspects 50 bottles and counts the number of defects, such as improper caps, labels, or damaged bottles. The defects observed over 10 consecutive days are recorded in the table below.

Day	1	2	3	4	5	6	7	8	9	10
Defects	3	4	2	5	3	6	7	3	4	2

Construct the c-chart to monitor the number of defects over the 10-day period. Identify any points that indicate the process is out of control.

Step 1: Average number of defects $\bar{c}$ $(k = 10)$

$$\begin{aligned}\bar{c} &= \frac{\sum c_i}{k} \\ &= \frac{3+4+2+5+3+6+7+3+4+2}{10} \\ &= \frac{39}{10} \\ &= \boxed{3.9}\end{aligned}$$

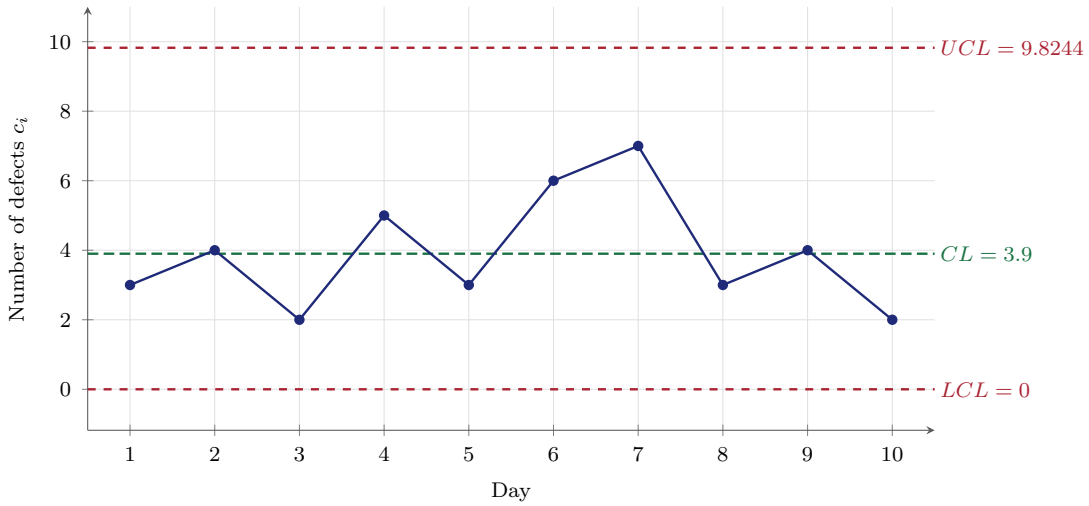
Step 2: Standard error $\sqrt{\bar{c}}$

$$\begin{aligned}\sigma_c &= \sqrt{\bar{c}} \\ &= \sqrt{3.9} \\ &= 1.9748\end{aligned}$$

Step 3: Control limits

$$\begin{aligned}UCL &= \bar{c} + 3\sqrt{\bar{c}} \\ &= 3.9 + 3(1.9748) \\ &= 3.9 + 5.9244 \\ &= \boxed{9.8244} \\ CL &= \bar{c} \\ &= \boxed{3.9} \\ LCL &= \bar{c} - 3\sqrt{\bar{c}} \\ &= 3.9 - 5.9244 \\ &= -2.0244 \\ &< 0 \\ \therefore\ LCL &= \boxed{0}\end{aligned}$$

Step 4: Control chart



Answer – process in control

All 10 points lie between $LCL = 0$ and $UCL = 9.8244$; the highest value is 7 (day 7).

The process is in control – only chance causes of variation are present.

* * *